

STA9794: A/B Testing & Causal Inference

CUNY Baruch

Differences-In-Differences (DiD) / Synthetic DiD

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1 Introduction

This document details my final project in STA9794 at CUNY Baruch. I used DiD / Synthetic DiD (SDiD) to estimate the causal effect on revenue of increasing product selection by 10% at a retailer. The data in this project are representative of supply chain data, but have been altered for anonymity.

2 Background

This project includes data from a three month pre-treatment period (Week 3-14) and a three month post-treatment period (Week 15-26). There is only one treatment store ("Store_1"), and ten control stores. The control stores are from the same geographic region as the treatment store, but they are far enough away so that customers would

Store	Week	Units_Sold	Revenue
Store_1	3	1,826	41,374.08
Store_1	4	1,558	35,692.89
Store_1	5	1,351	30,476.24
Store_1	6	1,703	38,933.86
Store_1	7	1,639	37,639.10
Store_1	8	1,552	35,544.75

not shop from multiple stores. In the analysis, I used a log transformation on units sold and revenue, as there are significant velocity differences between stores (e.g., one store does 4x the volume of another).

3 Treatment

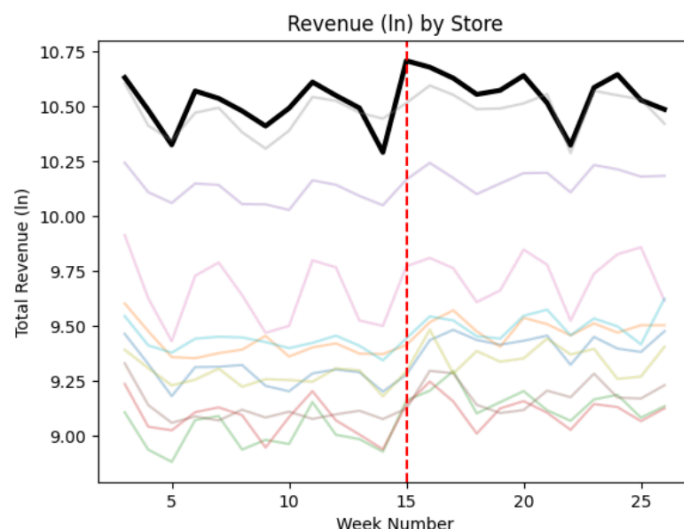
The treatment is an increase in the amount of products (+100 SKUs on a baseline of 1,000) at a store, in a new category. The other control stores did not receive these products. An increase in selection is a favorable outcome for a customer and one would expect a corresponding increase in revenue as customers would build bigger baskets. The treatment started in Week 15 at Store_1 and is assumed to be consistent with SUTVA (independent and consistent). The independence assumption is valid due to the geographic distance of the facilities and the treatment was consistent as all new products shipped on time, in full. A naive comparison that averages that

pre-treatment and post-treatment revenue and treatment and control stores shows the treatment store grew 40 bps faster (8.6% vs 8.2%), however this does not include any weighting.

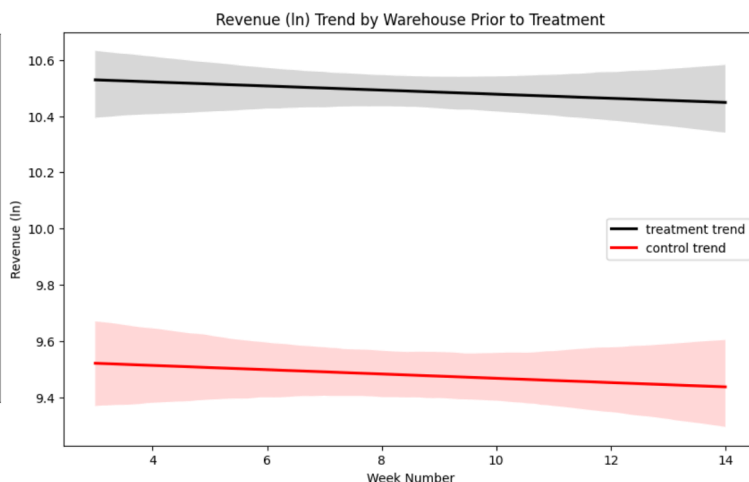
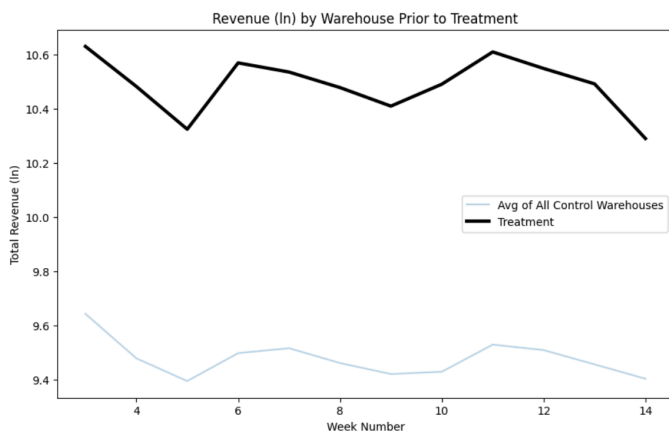
	Weekly Avg Revenue		
	Pre-Treatment	Post-Treatment	% Growth
Treatment	\$36,084	\$39,201	8.6%
Control	\$14,696	\$15,903	8.2%

4 EDA / Parallel Trends

The most important assumption for DiD is the parallel trends assumption. This assumes that the treatment and control stores move in the same direction. I validated this visually using time plots, as well as statistically by using an in-time placebo test. To the right are time plots showing the treatment store and control stores, before and after treatment, in terms of revenue and log-revenue.



To validate the parallel trends visually, I plotted the log-revenue of the treatment store and the mean of the control stores. I also plotted the trend in both, showing the data are consistent with the parallel trends assumption.



To statistically test the parallel trends assumption, I performed a Pre-Program Trend Test. I took the 12 weeks of pre-treatment data, and split it into a pseudo 6 weeks of pre-treatment and 6 weeks of post-treatment. I then ran a 2x2 DiD Regression. The placebo interaction coefficient was statistically insignificant ($p = .237$), which suggests no detectable differential pre-treatment trend during this period. However, there is only 6 weeks of data so the test has limited power.

OLS Regression Results						
Dep. Variable:	Revenue_In	R-squared:	0.313			
Model:	OLS	Adj. R-squared:	0.297			
Method:	Least Squares	F-statistic:	24.71			
Date:	Sun, 10 May 2026	Prob (F-statistic):	0.000135			
Time:	19:46:26	Log-Likelihood:	-76.185			
No. Observations:	132	AIC:	160.4			
Df Residuals:	128	BIC:	171.9			
Df Model:	3					
Covariance Type: cluster						
	coef	std err	z	P> z	[0.025	0.975]
Intercept	9.4992	0.148	64.087	0.000	9.209	9.790
treatment	1.0045	0.148	6.777	0.000	0.714	1.295
post_period	-0.0406	0.009	-4.460	0.000	-0.058	-0.023
treatment:post_period	0.0108	0.009	1.184	0.237	-0.007	0.029

5 2x2 DiD Regression

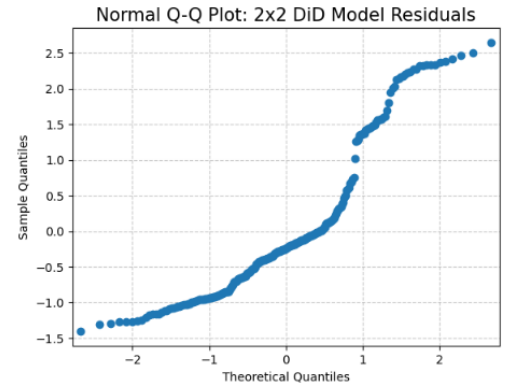
A 2x2 DiD design means there are two groups, treatment and control, for two periods, pre and post-treatment. In the below regression X represents a binary variable, 0/1 for control/treatment, which captures the baseline difference between the two groups. Y represents a binary variable, 0/1 for pre/post treatment, which captures the time trends shared by both groups.. XY represents the interaction variable which represents the causal effect of interest. It isolates the change in the treatment group in the post-treatment period compared to the control group during the same period. Below is the equation for the 2x2 DiD Regression.

$$Y_{it} = B_0 + B_1X_{it} + B_2Y_{it} + B_3XY_{it} + u_{it}$$

We observe the following issues with the 2x2 DiD Regression. The results show a statistically insignificant p-value for the interaction variable (.722). The errors appear to deviate from a normal distribution. A Durbin-Watson statistic of .161 shows strong positive autocorrelation, a natural consequence of the time series data. It also shows the following caution messages: covariance of constraints does not have full rank. We told the model to estimate three parameters: 1) treatment 2) time 3) interaction. The issue is

OLS Regression Results						
Dep. Variable:	Revenue_In	R-squared:	0.322			
Model:	OLS	Adj. R-squared:	0.315			
Method:	Least Squares	F-statistic:	24.86			
Date:	Mon, 11 May 2026	Prob (F-statistic):	0.000132			
Time:	00:29:04	Log-Likelihood:	-148.52			
No. Observations:	264	AIC:	305.0			
Df Residuals:	260	BIC:	319.4			
Df Model:	3					
Covariance Type: cluster						
	coef	std err	z	P> z	[0.025	0.975]
Intercept	9.4788	0.148	64.014	0.000	9.189	9.769
treatment	1.0099	0.148	6.820	0.000	0.720	1.300
post_period	0.0866	0.010	8.593	0.000	0.067	0.106
treatment:post_period	-0.0036	0.010	-0.356	0.722	-0.023	0.016
Omnibus:	37.251	Durbin-Watson:	0.161			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	49.755			
Skew:	1.060	Prob(JB):	1.57e-11			
Kurtosis:	3.177	Cond. No.	9.19			

we only have 11 clusters (1 treatment + 10 controls) and using a clustered standard of errors should be done with $G > 30$. This means the results are unreliable and we will move to Synthetic DiD using Placebo Permutation methods to handle inference with small samples.



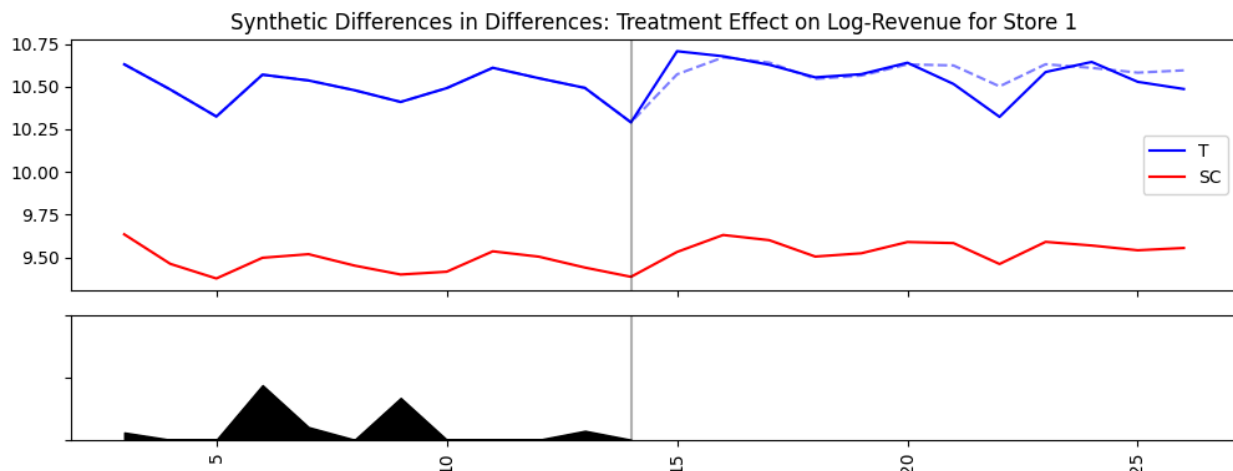
6 Synthetic DiD

Synthetic DiD (SDiD) uses Synthetic Control Methods (SCM) to construct a counterfactual of the treatment unit (Store_1). While DiD takes an equal weighting of all controls and all time periods, SDiD gives weighting to certain control stores (unit weight, ω) and time weight (λ) to match pre-treatment trends.

I used the SDiD implementation in [azcausal](#) from Amazon Science. I used the Placebo Permutation method to measure standard errors. This method assigns the treatment label to each control, one by one, to determine if the observed effect of the treatment unit (Store_1) is statistically significant. It is recommended where there is one treated unit in the sample (Abadie 2010).

The results show an estimated treatment effect of approximately -2.5%, meaning Store_1's log revenue declined by -2.5% compared to its synthetic control. However, since the confidence interval includes 0, we cannot say the results are statistically significant at the 5% level.

In the below plot, you can see the synthetic control in the blue-dotted line, compared to the solid blue line which represents the treated unit. Below that shows the weighting of the weeks chosen to represent a synthetic control that matches the pre-treatment trend of the treated unit. Please see the appendix for tables that show the weighting to construct the counterfactual.



y Conclusion

My biggest takeaway from this analysis is that a simple difference of means would show that the treatment unit's revenue grew 40 bps more than the control. However, this doesn't isolate the effects of the difference in the starting point from treatment vs control (e.g. Store_1's revenue is higher than average), and the time effects amongst all units. This may be the result of selection bias, Store_1 may have been chosen for the pilot because it generates the most revenue.

Once we run the 2x2 DiD Regression and the Synthetic DiD estimator, we see a decline in our effect of interest on revenue(ln), but the estimates are imprecise and the results do not provide strong evidence of a treatment effect. I ran the same experiments on units sold (ln) and found a statistically significant result of a -1.9% decline in units sold (ln) with SDiD, but an insignificant

one with the 2x2 DiD Regression. These results were noteworthy as they would suggest a mix into lower average sales priced products.

SDiD: Impact of Treatment on Revenue (ln)		
Panel		
Time Periods: 24 (12/12)		total (pre/post)
Units: 11 (10/1)		total (contr/treat)
ATT		
Effect (±SE): -0.025336 (±0.052905)		
Confidence Interval (95%): [-0.129028 , 0.078356]		(+/-)
Observed: 10.57		
Counter Factual: 10.60		
Percentage		
Effect (±SE): -0.239089 (±0.499243)		
Confidence Interval (95%): [-1.2176 , 0.739410]		(+/-)
Observed: 99.76		
Counter Factual: 100.00		
Cumulative		
Effect (±SE): -0.304036 (±0.634861)		
Confidence Interval (95%): [-1.5483 , 0.940268]		(+/-)
Observed: 126.86		
Counter Factual: 127.16		

There are a number of factors working against any potential claim to causality including the small sample size, a single treated unit, and possible non-random treatment assignment. There is also the possible operational complexity in handling additional products in a fixed capacity store leading to degradation in on time in full inventory management. While we cannot definitively say that offering a new product line caused an increase in revenue for the Retailer, a next step recommendation would be to expand the pilot to a few more stores and retest.

SDiD: Impact of Treatment on Units Sold (ln)		
Panel		
Time Periods: 24 (12/12)		total (pre/post)
Units: 11 (10/1)		total (contr/treat)
ATT		
Effect (±SE): -0.019798 (±0.005616)		
Confidence Interval (95%): [-0.030805 , -0.008791]		(-)
Observed: 7.4463		
Counter Factual: 7.4661		
Percentage		
Effect (±SE): -0.265170 (±0.075218)		
Confidence Interval (95%): [-0.412595 , -0.117744]		(-)
Observed: 99.73		
Counter Factual: 100.00		
Cumulative		
Effect (±SE): -0.237575 (±0.067391)		
Confidence Interval (95%): [-0.369658 , -0.105491]		(-)
Observed: 89.36		
Counter Factual: 89.59		

APPENDIX

Store	Weighting
Store_1	treatment
Store_2	12.8%
Store_3	12.8%
Store_4	9.0%
Store_5	6.9%
Store_6	14.6%
Store_7	9.6%
Store_8	9.2%
Store_9	9.4%
Store_10	9.7%
Store_11	6.1%
Total	100.0%

Week	Weighting
3	5.7%
4	0.0%
5	0.0%
6	43.6%
7	10.1%
8	0.0%
9	33.4%
10	0.0%
11	0.0%
12	0.0%
13	7.1%
14	0.0%
Total	100.0%